

Parle Tilak Vidyalaya Association's
Mulund College of Commerce, S.N.Road, Mulund (W), Mumbai-80.
T.Y.BBA Internal Examination, August 2025
Subject: **Optimization Techniques**

Date: 12/08/2025

Marks: 20

Time: 40 Min.

Q1. Solve the following assignment to maximize the revenue.

(05)

Jobs / Machines	J1	J2	J3	J4	J5
M1	7	5	9	8	11
M2	9	12	7	10	10
M3	8	5	4	6	9
M4	7	3	6	9	5
M5	1	1	1	1	1

Q2. Solve the following Assignment Problem to minimize the total cost. Also find which contractor is unsuccessful in getting a job

(05)

Roads / Contractors	R1	R2	R3	R4
C1	9	14	19	15
C2	7	17	20	19
C3	9	18	20	18
C4	10	12	18	19
C5	10	15	21	16

Q3. Given the following Transportation Problem with IFS. Check whether the solution is optimal, if not, find the optimum solution. Does the problem have an alternate solution

(05)

Distribution/ stores	D1	D2	D3	D4	Capacity
T1	10	8 (500)	7	12	500
T2	12 (50)	13 (50)	6 (450)	10 (50)	600
T3	8 (650)	10	12	14	650
Demand	700	550	450	300	

The figures in brackets are allocations.

Q4. Given the following table showing Profits. Find IFS using NWCR, LCM

(05)

Warehouses/ Factory	W1	W2	W3	W4	W5	Supply
F1	250	180	180	210	190	100
F2	200	220	230	200	240	125
F3	180	190	210	180	190	175
Demand	60	80	85	105	70	

